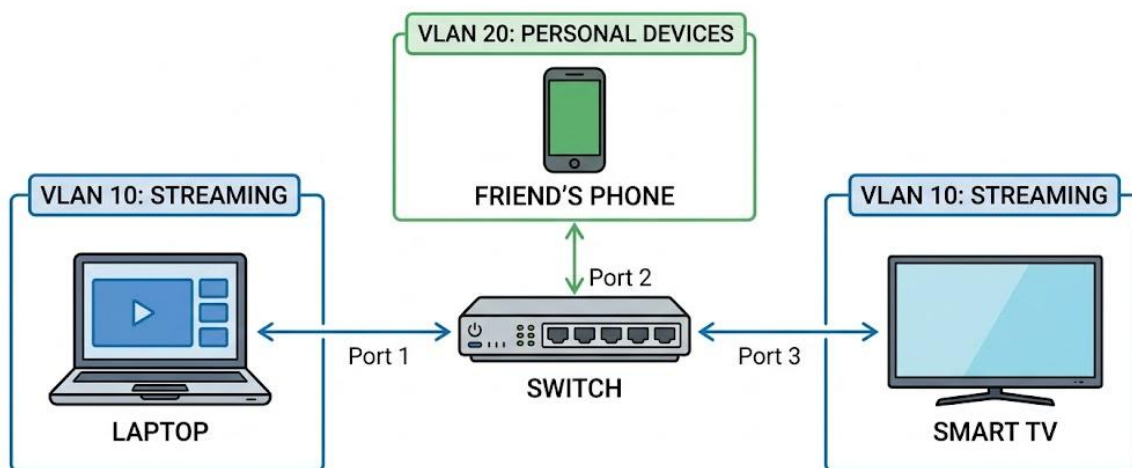


Switching and VLANs

Task 1 :- Draw a simple network diagram showing three devices (like your laptop, a friend's phone, and a smart TV) connected to a switch, and label which device is in which VLAN (e.g., VLAN 10 for streaming, VLAN 20 for personal devices).

ANS :-



Task 2 :- Configure a simulated switch using Cisco Packet Tracer or NetSim: create two VLANs (VLAN 10 and VLAN 20), assign three PCs to VLAN 10 and two PCs to VLAN 20, and verify that PCs in different VLANs cannot ping each other.

ANS :-

Step 1: Place Your Devices

- Drag 1 Switch (e.g., 2960) onto the screen.
- Drag 5 PCs onto the screen (Name them PC0 to PC4).
- Connect them with straight-through cables:
 - PC0, PC1, PC2 → Switch ports FastEthernet 0/1, 0/2, 0/3
 - PC3, PC4 → Switch ports FastEthernet 0/4, 0/5

Step 2: Assign IP Addresses

Click each PC → Desktop tab → IP Configuration:

- PC0: 192.168.1.1
- PC1: 192.168.1.2
- PC2: 192.168.1.3
- PC3: 192.168.1.4
- PC4: 192.168.1.5

(Set subnet mask for all to 255.255.255.0)

Step 3: Configure the Switch

Click the Switch → **CLI** tab and enter these commands:

```
enable
```

```
configure terminal
```

```
! Create the VLANs
```

```
vlan 10
```

```
name GroupA
```

```
vlan 20
```

```
name GroupB
```

```
exit
```

```
! Add PC0, PC1, PC2 to VLAN 10
```

```
interface range FastEthernet 0/1 - 3
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
exit
```

```
! Add PC3, PC4 to VLAN 20
```

```
interface range FastEthernet 0/4 - 5
```

```
switchport mode access
```

```
switchport access vlan 20
```

```
exit
```

Step 4: Verify It Works

1. Open PC0 (VLAN 10) → Desktop → Command Prompt.
2. Ping PC1 (VLAN 10): ping 192.168.1.2 → Success! (Same VLAN can talk)
3. Ping PC3 (VLAN 20): ping 192.168.1.4 → Request Timed Out! (Different VLANs blocked)

Task 3 :- In any network simulator, set up a scenario where you move a PC from VLAN 10 to VLAN 20. Show the before and after configuration commands and test connectivity to confirm the change.

ANS :-

Scenario Setup

- PC1 is connected to Switch port FastEthernet 0/1.
- IP Address: 192.168.1.1

Step 1: BEFORE Configuration :-

Initially, Port Fa0/1 is assigned to **VLAN 10**:

Plaintext

```
Switch(config)# interface FastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
```

- **Test:** PC1 can ping devices on VLAN 10, but **cannot** ping devices on VLAN 20.

Step 2: Move PC1 to VLAN 20 (AFTER Configuration)

Run this command on the switch to change the port's VLAN:

Plaintext

```
Switch(config)# interface FastEthernet 0/1
Switch(config-if)# switchport access vlan 20
```

Step 3: Test & Verify

1. Open Command Prompt on PC1.
2. Ping a VLAN 10 device: ping 192.168.1.2 → Request Timed Out! (No longer in VLAN 10)

3. Ping a VLAN 20 device: ping 192.168.1.4 → Reply Received! (Successfully moved to VLAN 20)
- **Check on Switch** : Run show vlan brief to confirm Fa0/1 now shows under VLAN 20.

Task 4 :- Write a short script or command list that would automate assigning 10 ports on a switch to VLAN 30, using the syntax of your chosen simulator (Packet Tracer, NetSim, or GNS3).

ANS :-

Plaintext

```
enable
configure terminal
vlan 30
  name TechTeam
interface range FastEthernet 0/1 - 10
  switchport mode access
  switchport access vlan 30
```

- ❓ **interface range Fa0/1 - 10** → Selects ports 1 through 10 all together.
- ❓ **switchport access vlan 30** → Moves all 10 ports into VLAN 30 instantly.